Table 1. Dictionary between category theory and linear algebra.

Category $\mathcal{C}$	Vector space V with a nondegenerate inner product
The set of morphisms $\mathrm{Hom}(X, Y)$	Inner product (x, y) on V (maybe nonsymmetric)
Opposite category $\mathcal{C}^{\mathrm{op}}$	Same space V with reversed inner product
The category **Sets**	The ground field k
Full subcategory in $\mathcal{C}$	Nondegenerate subspace in V
Functor $F : \mathcal{C} \to \mathcal{D}$	Linear operator $f : V \to W$
Functor $F : \mathcal{C} \to \mathbf{Sets}$	Linear functional $f \in V^* = \mathrm{Hom}(V, k)$
Representable functor	Linear functional $f \in V^*$ given by $f(v) = (u, v)$, $u \in V$
Yoneda lemma	Nondegeneracy of the inner product (on both sides)
Not all functors are representable	If $\dim V = \infty$, not $\forall f \in V^*$, $f(v) = (u, v)$
Left and right adjoint functors	Left and right adjoint operators
Adjoint functors don't always exist	Adjoint operators may not exist if $\dim V = \infty$
If they do, they are unique	If they do, they are unique
Left and right adjoints may not coincide	The inner product may be nonsymmetric

F, G are a pair of adjoint functors, then $F(X)$ represents the functor $Y \mapsto \mathrm{Hom}(X, G(Y))$ and $G(Y)$ represents the functor $X \mapsto \mathrm{Hom}(F(X), Y)$.

Remark 7.6.2. The terminology "left and right adjoint functors" is motivated by the analogy between categories and inner product spaces. More specifically, in Table 1 we have a useful dictionary between category theory and linear algebra, which helps to better understand many notions of category theory.